

Ch. # 7

MICROPROGRAMMED CONTROL

- Control Memory
- Address Sequencing
- Microprogram Example
- Design of Control Unit



Control Memory

- Control Unit
 - Initiate sequences of microoperations
 - Control signal (that specify microoperations) in a bus-organized system
 - groups of bits that select the paths in multiplexers, decoders, and arithmetic logic units
- Two major types of Control Unit
 - Hardwired Control : in Chap. 5
 - The control logic is implemented with gates, F/Fs, decoders, and other digital circuits
 - + Fast operation, - Wiring change(if the design has to be modified)
 - Microprogrammed Control : in this Chapter
 - The control information is stored in a control memory, and the control memory is programmed to initiate the required sequence of microoperations
 - + Any required change can be done by updating the microprogram in control memory,
 - - Slow operation
- Control Word

The control variables at any given time can be represented by a string of 1' s and 0' s.
- Microprogrammed Control Unit

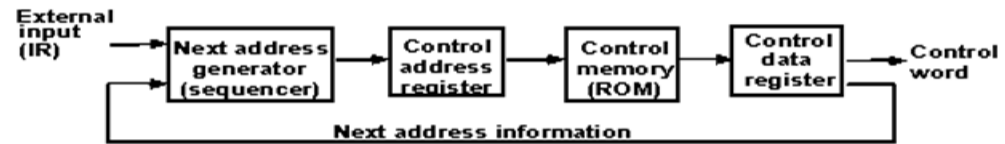
A control unit whose binary control variables are stored in memory (control memory)
- Control Memory: Storage in the mircoprogramed CU to store the microprogram.



- **Microinstruction** : *Control Word in Control Memory.*
 - The microinstruction specifies one or more microoperations.
- **Microprogram**
A sequence of microinstruction.
 - » **Dynamic microprogramming** : *Control Memory = RAM*
RAM can be used for writing (*to change a writable control memory*)
Microprogram is loaded initially from an auxiliary memory such as a magnetic disk
 - » **Static microprogramming** : *Control Memory = ROM*
Control words in ROM are made permanent during the hardware production.
- A computer that employs microprogrammed CU will have two types of memoris;
 1. Main Memory
 2. Control Memory



Microprogrammed Control Organization



- 1) Control Memory
 - A memory is part of a control unit : *Microprogram*
 - Computer Memory (*employs a microprogrammed control unit*)
 - » Main Memory : for storing user program (*Machine instruction/data*)
 - » Control Memory : for storing microprogram (*Microinstruction*)
- 2) Control Address Register
 - Specify the address of the microinstruction
- 3) Sequencer (= *Next Address Generator*)
 - Determine the address sequence that is read from control memory
 - Next address of the next microinstruction can be specified several way depending on the sequencer input
- 4) Control Data Register (= *Pipeline Register*)
 - Hold the microinstruction read from control memory
 - Allows the execution of the microoperations specified by the control word *simultaneously* with the generation of the next microinstruction



RISC Architecture Concept

- RISC(Reduced Instruction Set Computer) system use hardwired control rather than microprogrammed control : **Sec. 8-8**

